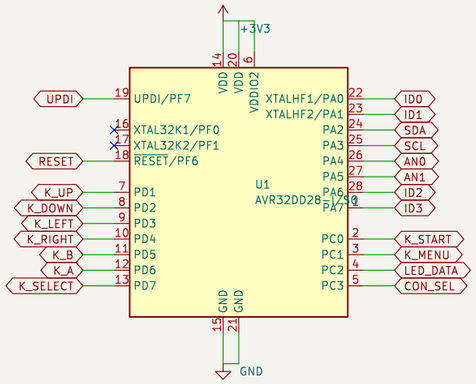
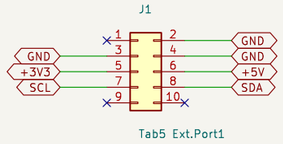
MCU — AVR32DD28 (SOIC-28) running firmware_avrdd (I2C client 0x6D, RetroPad Extended)

Tab5 Ext.Port1 (M5Stack Tab5 Keyboard P1 pinout) — verify pin 1 orientation before fab

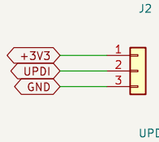
Buttons — one MCU pin each, switch to GND (internal pull-up); slot table in firmware_avrdd/pinmap.h

PD1–PD7, PC0–PC3, PF0–PF1: one button per pin (internal pull-ups)

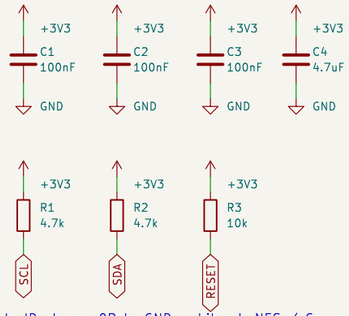


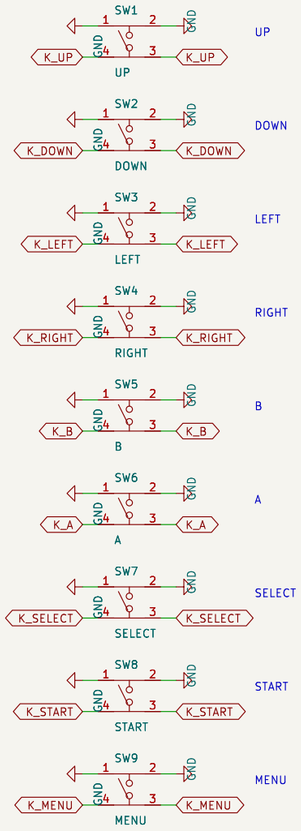


UPDI programming (SerialUPDI / MPLAB Snap)

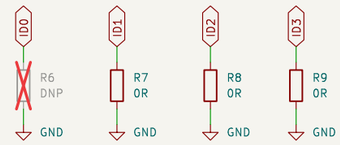


Decoupling (VDD x2, VDDIO2), I2C pull-ups, RESET pull-up

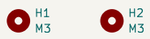


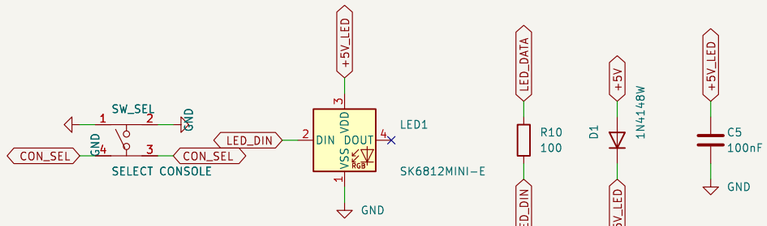


Console ID straps: OR to GND = bit set. NES / Game Boy / Master System: straps = ID 14 (fit R7, R8, R9), console picked with SW_SEL (NES / GB / SMS)



Mounting holes (M3, 96 mm apart)





Console select: SW_SEL steps NES / GB / SMS (PC3); LED1 shows it (PC2 via R10). LED1 runs from J1's 5 V through D1 (~4.3 V)

Tab5 keyboard-port board: AVR32DD28 + firmware_avrdd, I2C 0x6D
Generated by kicad_gen/gen_sch.py from gen_pcb.py and layout.py
RetroESP32-P4

Sheet: /
File: retropad_nes_gb_sms.kicad_sch

Title: RetroPad NES / Game Boy / Master System controller (console IDs 1/

Size: A3	Date: 2026-10-05	**Rev: 0.1**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1